Setup and Deployment Guide

Run it locally, push to GitHub, publish the dashboard, and publish to Power BI and Tableau Public

Compliance Audit Data Automation | Rahul Mistari

What you will need before starting

- Python 3.9 or later, installed and on your system PATH.
- Git, installed and configured with your name and email.
- A free GitHub account, for hosting the code and publishing the dashboard.
- Tableau Public Desktop (free), only if you want to publish there. Optional.
- Power BI Desktop (free, Windows only), only if you want to publish there. Optional.

All commands below are typed into a terminal (Command Prompt, PowerShell, or macOS/Linux Terminal), inside the project folder unless stated otherwise.

Step 1: Get the project onto your computer

If you were given a .zip file, unzip it anywhere convenient, then open a terminal in that folder.

If you are starting from an existing GitHub repository instead, clone it:

```
git clone <your-repo-url>
cd compliance-audit-data-automation
```

Step 2: Run it locally

Create and activate a virtual environment (optional but recommended), then install the required packages:

```
python -m venv .venv
source .venv/bin/activate  # on Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

Generate the synthetic sample data:

```
python generate_sample_data.py
```

Run the full validation workflow. This writes the audit trail and summary to the outputs folder:

```
python -m src.main
```

Run the test suite to confirm everything works:

```
pytest -v
```

Build the Tableau Public / Power BI ready export files:

```
python build_bi_exports.py
```

At this point the project is fully working on your machine, including the outputs folder, the data/bi_exports folder, and the dashboard file at docs/index.html, which you can open directly in a browser to preview it.

Step 3: Create a GitHub repository and push the code

On github.com, sign in, click the plus icon in the top right, and choose New repository.

Give it a name (for example compliance-audit-data-automation), leave it public if you want the dashboard to be publicly viewable, and do not initialize it with a README, license, or .gitignore, since the project already has these.

Click Create repository. GitHub will show you a remote URL, looking like https://github.com/your-username/compliance-audit-data-automation.git

Back in your terminal, inside the project folder, set your Git identity if you have not already, then push:

```
git config user.name "Your Name"
git config user.email "your-email@example.com"
git remote add origin
https://github.com/your-username/compliance-audit-data-automation.git
git push -u origin main
```

If the repository already has commit history with a different remote configured, or if you are re-pushing rewritten history, use git push -u origin main --force instead. Force-pushing overwrites whatever was previously on GitHub, so only do this if that is what you intend.

Refresh the repository page on GitHub. You should see all the project files, including the src, docs, tests, and data folders.

Step 4: Publish the dashboard with GitHub Pages

On your repository's GitHub page, click Settings.

In the left sidebar, click Pages.

Under Build and deployment, set Source to Deploy from a branch.

Set Branch to main, and the folder to /docs, then click Save.

Wait one to two minutes. Refresh the page. GitHub will show a message with your live URL, in the form:
https://your-username.github.io/compliance-audit-data-automation/

Open that URL. You should see the results dashboard, with tabs for Results, What do these rules mean, Rules by country, and Where this data comes from.

The same URL plus a filename gives direct links to the two PDF companions, for example:
https://your-username.github.io/compliance-audit-data-automation/Project_Explainer.pdf
https://your-username.github.io/compliance-audit-data-automation/Why_These_Compliance_Rules_Exist.pdf

Step 5: Add the dashboard link to your portfolio

Copy the GitHub Pages URL from Step 4.

In your portfolio site or resume, add it as a project link, for example labeled "Live dashboard" or "View results."

Optional: update the placeholder link near the top of this project's README.md with the same URL, so anyone reading the repository can find it too.

Step 6: Publish to Tableau Public

Install Tableau Public Desktop (free) from Tableau's website, if you have not already.

Open it. Choose Connect, then Text File (for a .csv) or Microsoft Excel (for a .xlsx).

Select data/bi_exports/audit_results.csv or data/bi_exports/compliance_audit_workbook.xlsx from the project folder.

Create a new worksheet. Drag Applies To onto Columns and Result onto Rows, then choose a bar chart. This recreates the "Findings by domain" chart from the HTML dashboard.

Create a second worksheet. Drag Result onto Color and choose a pie chart, for the pass/fail split.

Create a third worksheet as a simple table, filtered to Needs Review equals Yes, for the escalation list.

Create a new Dashboard and drag all three worksheets onto it.

Go to File, then Save to Tableau Public. Sign in, or create a free account if you do not have one.

Tableau Public publishes the workbook and gives you a public URL. Copy that URL for your portfolio.

Step 7: Publish to Power BI

Install Power BI Desktop (free, Windows only) from Microsoft's website, if you have not already.

Open it. Go to Home, then Get Data, then choose Text/CSV or Excel workbook.

Select data/bi_exports/audit_results.csv or data/bi_exports/compliance_audit_workbook.xlsx, then click Load.

Add a Card visual for total checks, a Stacked Column Chart (Applies To on the axis, Result as the legend), and a Donut Chart for the pass/fail split.

Add a Table visual filtered to Needs Review equals Yes, for the escalation list.

Go to File, then Publish, then Publish to Power BI. This requires a free Power BI account and publishes the report to a workspace in the Power BI service.

For a public link similar to Tableau Public, open the report in the Power BI service in your browser, go to File, then Publish to web. This generates a public, embeddable URL. Copy it for your portfolio.

Where each link ends up

Portfolio item	**Where the link comes from**
Live dashboard	GitHub Pages URL, from Step 4
Source code	The GitHub repository URL itself, from Step 3
Project explainer (PDF)	GitHub Pages URL plus /Project_Explainer.pdf
Why these rules exist (PDF)	GitHub Pages URL plus /Why_These_Compliance_Rules_Exist.pdf
Tableau Public dashboard	The public URL from Step 6
Power BI report	The published-to-web URL from Step 7

Notes and troubleshooting

If git push is rejected because the remote already has commits you do not have locally, and you are certain you want your local version to replace what is on GitHub, use git push -u origin main --force. This permanently overwrites the remote history.

Commits only show up as yours on GitHub, with your avatar and profile linked, if the commit's author email matches a verified email on your GitHub account. Set it with git config user.email before committing.

To refresh the sample data and results with a new random run, repeat Step 2, then commit and push the changed files in data, outputs, and data/bi_exports.

Neither Tableau Public nor Power BI accept an uploaded dashboard file directly from outside their own applications. Both require opening the data in the desktop app and publishing from there, which is the normal workflow for each platform, not a limitation specific to this project.

Quick command reference

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python generate_sample_data.py
python -m src.main
pytest -v
python build_bi_exports.py
git add -A
git commit -m "your message"
git push -u origin main
```